

2 Monomials—Laws of Indices—Degree

Exercises II

1. Simplify—

$$\frac{5^7 \times 12^4 \times 32^2 \times (3^2 \times 4^2 \times 5)^2}{(3 \times 15 \times 2^3)^{10}}.$$

First, we simplify the numerator,

$$\begin{aligned} & 5^7 \times 12^4 \times 32^2 \times (3^2 \times 4^2 \times 5)^2 \\ &= 5^7 \times (2 \times 2 \times 3)^4 \times (2^5)^2 \times (3^2 \times (2 \times 2)^2 \times 5)^2 && \text{(arithmetical factorization)} \\ &= 5^7 \times 2^4 \times 2^4 \times 3^4 \times 2^{10} \times 3^4 \times 2^4 \times 2^4 \times 5^2 && ((ab)^n = a^n b^n; (a^m)^n = a^{mn}) \\ &= 5^9 \times 3^8 \times 2^{26} && \text{(commutation; } a^m a^n = a^{m+n}) \end{aligned}$$

Next, we simplify the denominator,

$$\begin{aligned} & (3 \times 15 \times 2^3)^{10} \\ &= 3^{10} \times 15^{10} \times 2^{30} && ((ab)^n = a^n b^n; (a^m)^n = a^{mn}) \\ &= 3^{10} \times (3 \times 5)^{10} \times 2^{30} && \text{(arithmetical factorization)} \\ &= 3^{10} \times 3^{10} \times 5^{10} \times 2^{30} && ((ab)^n = a^n b^n) \\ &= 3^{20} \times 5^{10} \times 2^{30} && (a^m a^n = a^{m+n}) \end{aligned}$$

Finally, we simplify the fraction,

$$\begin{aligned} & \frac{5^9 \times 3^8 \times 2^{26}}{3^{20} \times 5^{10} \times 2^{30}} \\ &= \frac{1}{3^{12} \times 2^4 \times 5} && \left(\frac{a^m}{a^n} = \frac{1}{a^{n-m}}; n > m\right) \\ &= \frac{1}{42515280} \end{aligned}$$

2. Which is greater, $(2^2)^{2^2}$ or 2^{2^2} ? Find the difference between them.

Observe that $2^{2^{2^2}} = 2^{2^4} = 2^{16} = 65536$, whereas $(2^2)^{2^2} = (2^2)^4 = 2^8 = 256$.
Therefore $2^{2^{2^2}}$ is larger.

3. Simplify—

$$\frac{2^{2^2}}{2(2^2)^2}.$$

$$\frac{2^{2^2}}{2(2^2)^2} = \frac{(2 \times 2)^2}{2 \times (2 \times 2)^2} = \frac{1}{2}$$

4. Simplify—

$$\frac{36^2 a^7 b^5 c^3 d^2}{81 a^4 b^3 c^3}.$$

$$\frac{36^2 a^7 b^5 c^3 d^2}{81 a^4 b^3 c^3} = \frac{(9 \times 4)^2 a^7 b^5 c^3 d^2}{(9 \times 9) a^4 b^3 c^3} = \frac{9^2 \times 4^2 \times a^7 b^5 c^3 d^2}{9^2 a^4 b^3 c^3} = 4^2 a^3 b^2 d^2 = 16 a^3 b^2 d^2$$

5. Express in its simplest form—

$$\left(\frac{a^4 c^3 y^2}{a^3 b^2 c^3 x^2 y^2} \right)^2 \times \left(\frac{a^3 b^3 x^3}{a^3 b^2 c^3 x^3 y^2} \right)^3 \times \left(\frac{b^2 c^4 x^2 y}{a^4 b^3 x^3 y^3} \right)^4.$$

$$\begin{aligned}
& \left(\frac{a^4 c^3 y^2}{a^3 b^2 c^3 x^2 y^2} \right)^2 \times \left(\frac{a^3 b^3 x^3}{a^3 b^2 c^3 x^3 y^2} \right)^3 \times \left(\frac{b^2 c^4 x^2 y}{a^4 b^3 x^3 y^3} \right)^4 \\
&= \left(\frac{a}{b^2 x^2} \right)^2 \times \left(\frac{b}{c^3 y^2} \right)^3 \times \left(\frac{b^2 c^4 x^2 y}{a^4 b^3 x^3 y^3} \right)^4 & \left(\frac{a^m}{a^n} = a^{m-n}; m > n \right) \\
&= \left(\frac{a}{b^2 x^2} \right)^2 \times \left(\frac{b}{c^3 y^2} \right)^3 \times \left(\frac{c^4}{a^4 b x y^2} \right)^4 & \left(\frac{a^m}{a^n} = \frac{1}{a^{n-m}}; n > m \right) \\
&= \frac{(a)^2}{(b^2 x^2)^2} \times \frac{(b)^3}{(c^3 y^2)^3} \times \frac{(c^4)^4}{(a^4 b x y^2)^4} & \left(\left(\frac{a}{b} \right)^m = \frac{a^m}{b^m} \right) \\
&= \frac{a^2}{(b^2)^2 (x^2)^2} \times \frac{b^3}{(c^3)^3 (y^2)^3} \times \frac{(c^4)^4}{(a^4)^4 (y^2)^4 b^4 x^4} & ((ab)^m = a^m b^m) \\
&= \frac{a^2}{b^4 x^4} \times \frac{b^3}{c^9 y^6} \times \frac{c^{16}}{a^{16} y^8 b^4 x^4} & ((a^m)^n = a^{mn}) \\
&= \frac{a^2 b^3 c^{16}}{b^4 x^4 c^9 y^6 a^{16} y^8 b^4 x^4} \\
&= \frac{a^2 b^3 c^{16}}{b^8 x^8 c^9 y^{14} a^{16}} & (a^m a^n = a^{m+n}) \\
&= \frac{c^{16}}{b^5 x^8 c^9 y^{14} a^{14}} & \left(\frac{a^m}{a^n} = \frac{1}{a^{n-m}}; n > m \right) \\
&= \frac{c^7}{b^5 x^8 y^{14} a^{14}} & \left(\frac{a^m}{a^n} = a^{m-n}; m > n \right)
\end{aligned}$$

6. Simplify—

$$\begin{aligned}
& \left(\frac{45a^3 b^2 c^4}{27a^2 b^2 c} \right)^2 \times \left(\frac{243a^4 b^4 c^4 x}{180a^2 b c} \right)^2 \\
&= \left(\frac{(9 \times 5)a^3 b^2 c^4}{(9 \times 3)a^2 b^2 c} \right)^2 \times \left(\frac{3^5 a^4 b^4 c^4 x}{(2^2 \times 3^2 \times 5)a^2 b c} \right)^2 \\
&= \left(\frac{5ac^3}{3} \right)^2 \times \left(\frac{3^3 a^2 b^3 c^3 x}{2^2 \times 5} \right)^2 \\
&= \frac{5^2 a^2 c^6}{3^2} \times \frac{3^6 a^4 b^6 c^6 x^2}{2^4 \times 5^2} \\
&= \frac{3^4 a^6 b^6 c^{12} x^2}{2^4} \\
&= \frac{81a^6 b^6 c^{12} x^2}{16}
\end{aligned}$$

7. Simplify—

$$(3xyz)^2 \times \frac{x^6 y^9}{x^9 y^6} \div \left(\frac{3x^2 z^2}{y^2 z^2} \right)^2.$$

$$\begin{aligned} (3xyz)^2 \times \frac{x^6 y^9}{x^9 y^6} &\div \left(\frac{3x^2 z^2}{y^2 z^2} \right)^2 \\ &= 9x^2 y^2 z^2 \times \frac{y^3}{x^3} \div \frac{9x^4}{y^4} \\ &= \frac{9y^5 z^2}{x} \div \frac{9x^4}{y^4} \\ &= \frac{y^9 z^2}{x^5} \end{aligned}$$

8. Simplify—

$$\frac{(x^3 y^2 z^2)^7 \times (y^3 z^2 x^2)^7 \times (z^3 x^2 y^2)^7}{\left(\frac{x^2}{yz} \right)^7 \times \left(\frac{y^2}{zx} \right)^7 \times \left(\frac{z^2}{xy} \right)^7}.$$

$$\begin{aligned} &\frac{(x^3 y^2 z^2)^7 \times (y^3 z^2 x^2)^7 \times (z^3 x^2 y^2)^7}{\left(\frac{x^2}{yz} \right)^7 \times \left(\frac{y^2}{zx} \right)^7 \times \left(\frac{z^2}{xy} \right)^7} \\ &= \left(\frac{x^3 y^2 z^2 y^3 z^2 x^2 z^3 x^2 y^2}{\frac{x^2}{yz} \times \frac{y^2}{zx} \times \frac{z^2}{xy}} \right)^7 && ((ab)^m = a^m b^m; \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}) \\ &= \left(\frac{x^7 y^7 z^7}{\frac{x^2}{yz} \times \frac{y^2}{zx} \times \frac{z^2}{xy}} \right)^7 && (a^m a^n = a^{m+n}) \\ &= \left(\frac{x^7 y^7 z^7}{\frac{x^2 y^2 z^2}{yzzxxy}} \right)^7 \\ &= \left(\frac{x^7 y^7 z^7}{\frac{x^2 y^2 z^2}{y^2 z^2 x^2}} \right)^7 && (\text{definition of exponent}) \\ &= (x^7 y^7 z^7)^7 \\ &= (xyz)^{49} && ((a^m)^n = a^{mn}) \end{aligned}$$

9. Simplify—

$$\left(\frac{x^m}{x^n} \right)^{m+n} \times \left(\frac{x^n}{x^l} \right)^{n+l} \times \left(\frac{x^l}{x^m} \right)^{l+m}.$$

$$\begin{aligned}
& \left(\frac{x^m}{x^n}\right)^{m+n} \times \left(\frac{x^n}{x^l}\right)^{n+l} \times \left(\frac{x^l}{x^m}\right)^{l+m} \\
& \quad \frac{x^{m^2+mn}}{x^{n^2+mn}} \times \frac{x^{n^2+nl}}{x^{l^2+nl}} \times \frac{x^{l^2+lm}}{x^{m^2+lm}} \quad ((a^m)^n = a^{mn}; (\frac{a}{b})^m = \frac{a^m}{b^m}) \\
& = \frac{x^{m^2} x^{mn}}{x^{n^2} x^{mn}} \times \frac{x^{n^2} x^{nl}}{x^{l^2} x^{nl}} \times \frac{x^{l^2} x^{lm}}{x^{m^2} x^{lm}} \quad (a^m a^n = a^{m+n}) \\
& \quad \quad \quad = 1 \quad (\frac{a}{a} = 1)
\end{aligned}$$

10. Simplify—

$$\left\{ \frac{(a^{p-q})^q \times (a^{q-r})^{r-p}}{(a^{r+q})^{r-q}} \right\}^{1/r}.$$

$$\begin{aligned}
& \left\{ \frac{(a^{p-q})^q \times (a^{q-r})^{r-p}}{(a^{r+q})^{r-q}} \right\}^{1/r} = \left\{ \frac{a^{pq-q^2} \times a^{qr-qp-r^2+rp}}{a^{r^2-q^2}} \right\}^{1/r} \\
& = \left\{ \frac{a^{pq-q^2+qr-qp-r^2+rp}}{a^{r^2-q^2}} \right\}^{1/r} = \left\{ \frac{a^{r(q+p)} a^{-r^2} a^{-q^2}}{a^{r^2} a^{-q^2}} \right\}^{1/r} \\
& \quad \quad \quad = \frac{a^{q+p} a^{-r}}{a^r} = a^{q+p-2r}
\end{aligned}$$

11. Simplify—

$$\frac{(x^{a-b} \times x^{b-c})^a \times \left(\frac{x^a}{x^c}\right)}{(x^a \times x^c)^a \div (x^{a+c})^c}.$$

First, the numerator,

$$\begin{aligned}
& (x^{a-b} \times x^{b-c})^a \times \left(\frac{x^a}{x^c}\right) \\
& = (x^{a-c})^a \times x^{a-c} \quad (a^m a^n = a^{m+n}; \frac{a^m}{a^n} = a^{m-n}) \\
& = x^{a(a-c)} \times x^{a-c} \quad ((a^m)^n = a^{mn}) \\
& = x^{(a-c)(a+1)} \quad (a^m a^n = a^{m+n})
\end{aligned}$$

Next, the denominator,

$$\begin{aligned}
& (x^a \times x^c)^a \div (x^{a+c})^c \\
& = (x^{a+c})^a \div (x^{a+c})^c \quad (a^m a^n = a^{m+n}) \\
& = x^{a(a+c)} \div x^{c(a+c)} \quad ((a^m)^n = a^{mn}) \\
& = x^{(a+c)(a-c)} \quad (a^m \div a^n = a^{m-n})
\end{aligned}$$

Finally,

$$\frac{x^{(a-c)(a+1)}}{x^{(a-c)(a+c)}} = x^{(a-c)(a+1)-(a-c)(a+c)} = x^{(a-c)(a+1-a-c)} = x^{(a-c)(1-c)}.$$

12. Simplify—

$$\left(\frac{x^p}{x^q}\right)^{p+q} \div \left(\frac{x^{p+q}}{x^{p-q}}\right)^{p^2/q}.$$

$$\begin{aligned} & \left(\frac{x^p}{x^q}\right)^{p+q} \div \left(\frac{x^{p+q}}{x^{p-q}}\right)^{p^2/q} \\ &= (x^{p-q})^{p+q} \div (x^{2q})^{p^2/q} & \left(\frac{a^m}{a^n} = a^{m-n}\right) \\ &= x^{p^2-q^2} \div x^{2p^2} & ((a^m)^n = a^{mn}) \\ &= \frac{1}{x^{p^2+q^2}} & (a^m \div a^n = \frac{1}{a^{n-m}}; n > m) \end{aligned}$$

13. Simplify—

$$\left\{ \left(\frac{x^l}{x^m}\right)^l \times \left(\frac{x^m}{x^l}\right)^m \right\} \div \{(x^l)^l \times (x^m)^m\} \times \{(x^m)^l \times (x^l)^m\}.$$

First, we simplify each term inside brackets,

$$\begin{aligned} \left(\frac{x^l}{x^m}\right)^l \times \left(\frac{x^m}{x^l}\right)^m &= x^{l(l-m)} \times x^{m(m-l)} = x^{l^2+m^2-2lm}, \\ (x^l)^l \times (x^m)^m &= x^{l^2+m^2}, \\ (x^m)^l \times (x^l)^m &= x^{2lm}. \end{aligned}$$

Then, we perform the division and multiplication chain,

$$\begin{aligned} x^{l^2+m^2-2lm} \div x^{l^2+m^2} \times x^{2lm} &= x^{l^2+m^2-2lm} \times x^{2lm} \div x^{l^2+m^2} \\ &= x^{l^2+m^2} \div x^{l^2+m^2} & (a^m a^n = a^{m+n}) \\ &= 1. & \left(\frac{a}{a} = 1\right) \end{aligned}$$

14. Prove that—

$$\frac{(yz)^{qr}(zx)^{rp}(xy)^{pq}}{(y^{q-1}z^{r-1})^p(z^{r-1}x^{p-1})^q(x^{p-1}y^{q-1})^r} = \frac{(xyz)^{p+q+r}}{x^p y^q z^r}.$$

Observe that,

$$\begin{aligned}
& \frac{(yz)^{qr}(zx)^{rp}(xy)^{pq}}{(y^{q-1}z^{r-1})^p(z^{r-1}x^{p-1})^q(x^{p-1}y^{q-1})^r} \\
&= \frac{y^{qr}z^{qr}z^{rp}x^{rp}x^{pq}y^{pq}}{y^{pq-p}z^{pr-p}z^{qr-q}x^{pq-q}x^{pr-r}y^{qr-r}} & ((ab)^n = a^n b^n; (a^m)^n = a^{mn}) \\
&= \frac{y^{qr}z^{qr}z^{rp}x^{rp}x^{pq}y^{pq}}{y^{pq}y^{-p}z^{pr}z^{-p}z^{qr}z^{-q}x^{pq}x^{-q}x^{pr}x^{-r}y^{qr}y^{-r}} & (a^{m+n} = a^m a^n) \\
&= \frac{1}{y^{-p}z^{-p}z^{-q}x^{-q}x^{-r}y^{-r}} & (\frac{a^m}{a^m} = 1) \\
&= \frac{1}{y^{-p}z^{-p}z^{-q}x^{-q}x^{-r}y^{-r}} \times 1 \\
&= \frac{1}{y^{-p}z^{-p}z^{-q}x^{-q}x^{-r}y^{-r}} \times \frac{x^p x^q x^r y^p y^q y^r z^p z^q z^r}{x^p x^q x^r y^p y^q y^r z^p z^q z^r} & (\frac{a}{a} = 1) \\
&= \frac{x^p x^q x^r y^p y^q y^r z^p z^q z^r}{(y^{-p}z^{-p}z^{-q}x^{-q}x^{-r}y^{-r})(x^p x^q x^r y^p y^q y^r z^p z^q z^r)} \\
&= \frac{x^p x^q x^r y^p y^q y^r z^p z^q z^r}{(y^{-p+p}z^{-p+p}z^{-q+q}x^{-q+q}x^{-r+r}y^{-r+r})(x^p y^q z^r)} & (a^m a^n = a^{m+n}) \\
&= \frac{x^p x^q x^r y^p y^q y^r z^p z^q z^r}{x^p y^q z^r} & (a^{-m} a^m = 1) \\
&= \frac{x^{p+q+r} y^{p+q+r} z^{p+q+r}}{x^p y^q z^r} & (a^m a^n = a^{m+n}) \\
&= \frac{(xyz)^{p+q+r}}{x^p y^q z^r} & (a^n b^n c^n = (abc)^n)
\end{aligned}$$

15. Distribute the product—

$$(x^{b-c} - x^{c-a} + x^{a-b}) \left(\frac{1}{x^{b-c}} + \frac{1}{x^{c-a}} + \frac{1}{x^{a-b}} \right).$$

$$\begin{aligned}
& \frac{x^{b-c}}{x^{b-c}} + \frac{x^{b-c}}{x^{c-a}} + \frac{x^{b-c}}{x^{a-b}} - \frac{x^{c-a}}{x^{b-c}} - \frac{x^{c-a}}{x^{c-a}} - \frac{x^{c-a}}{x^{a-b}} + \frac{x^{a-b}}{x^{b-c}} + \frac{x^{a-b}}{x^{c-a}} + \frac{x^{a-b}}{x^{a-b}} \\
&= 1 + x^{(b-c)-(c-a)} + x^{(b-c)-(a-b)} - x^{(c-a)-(b-c)} - 1 - x^{(c-a)-(a-b)} \\
&\quad + x^{(a-b)-(b-c)} + x^{(a-b)-(c-a)} + 1 \\
&= 1 + x^{a+b-2c} + x^{-a+2b-c} - x^{-a-b+2c} - x^{-2a+b+c} + x^{a-2b+c} + x^{2a-b-c} \\
&= 1 + x^{2a-b-c} - x^{-2a+b+c} + x^{-a+2b-c} + x^{a-2b+c} - x^{-a-b+2c} + x^{a+b-2c}
\end{aligned}$$

16. Distribute—

$$\left(a^p + \frac{1}{b^q} \right) \left(a^q + \frac{1}{b^p} \right)^2.$$

$$\begin{aligned}
\left(a^p + \frac{1}{b^q}\right) \left(a^q + \frac{1}{b^p}\right)^2 &= \left(a^p + \frac{1}{b^q}\right) \left(a^q + \frac{1}{b^p}\right) \left(a^q + \frac{1}{b^p}\right) \\
&= \left(a^{p+q} + \frac{a^p}{b^p} + \frac{a^q}{b^q} + \frac{1}{b^{q+p}}\right) \left(a^q + \frac{1}{b^p}\right) \\
&= a^{p+2q} + \frac{a^{p+q}}{b^p} + \frac{a^{p+q}}{b^p} + \frac{a^p}{b^{2p}} + \frac{a^{2q}}{b^q} + \frac{a^q}{b^{q+p}} + \frac{a^q}{b^{q+p}} + \frac{1}{b^{q+2p}} \\
&= a^{p+2q} + \frac{2a^{p+q}}{b^p} + \frac{a^p}{b^{2p}} + \frac{a^{2q}}{b^q} + \frac{2a^q}{b^{q+p}} + \frac{1}{b^{q+2p}}
\end{aligned}$$

17. If $m = a^x$, $n = a^y$, $a^2 = (m^y n^x)^z$; show that $xyz = 1$.

Substituting $m = a^x$ and $n = a^y$ in the third relation,

$$\begin{aligned}
a^2 &= \{(a^x)^y (a^y)^x\}^z \\
&= \{a^{xy} \times a^{xy}\}^z && ((a^m)^n = a^{mn}) \\
&= (a^{2xy})^z && (a^m a^n = a^{m+n}) \\
&= a^{2xyz}. && ((a^m)^n = a^{mn})
\end{aligned}$$

Since the two powers of a are equal, their indices are equal, therefore $xyz = 1$.